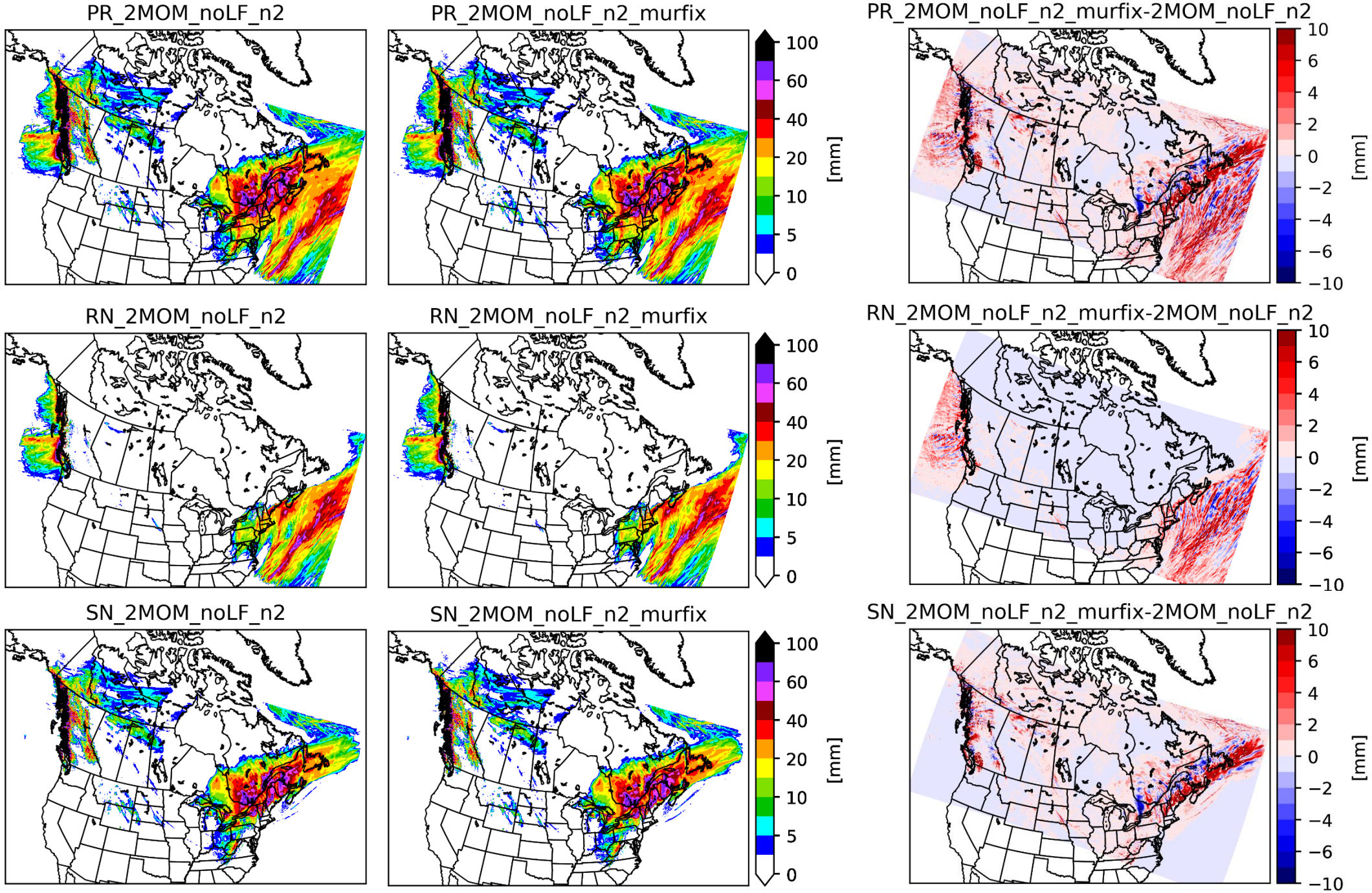
Problem of increase PR and SN with nCat=2,3…

Melissa Cholette

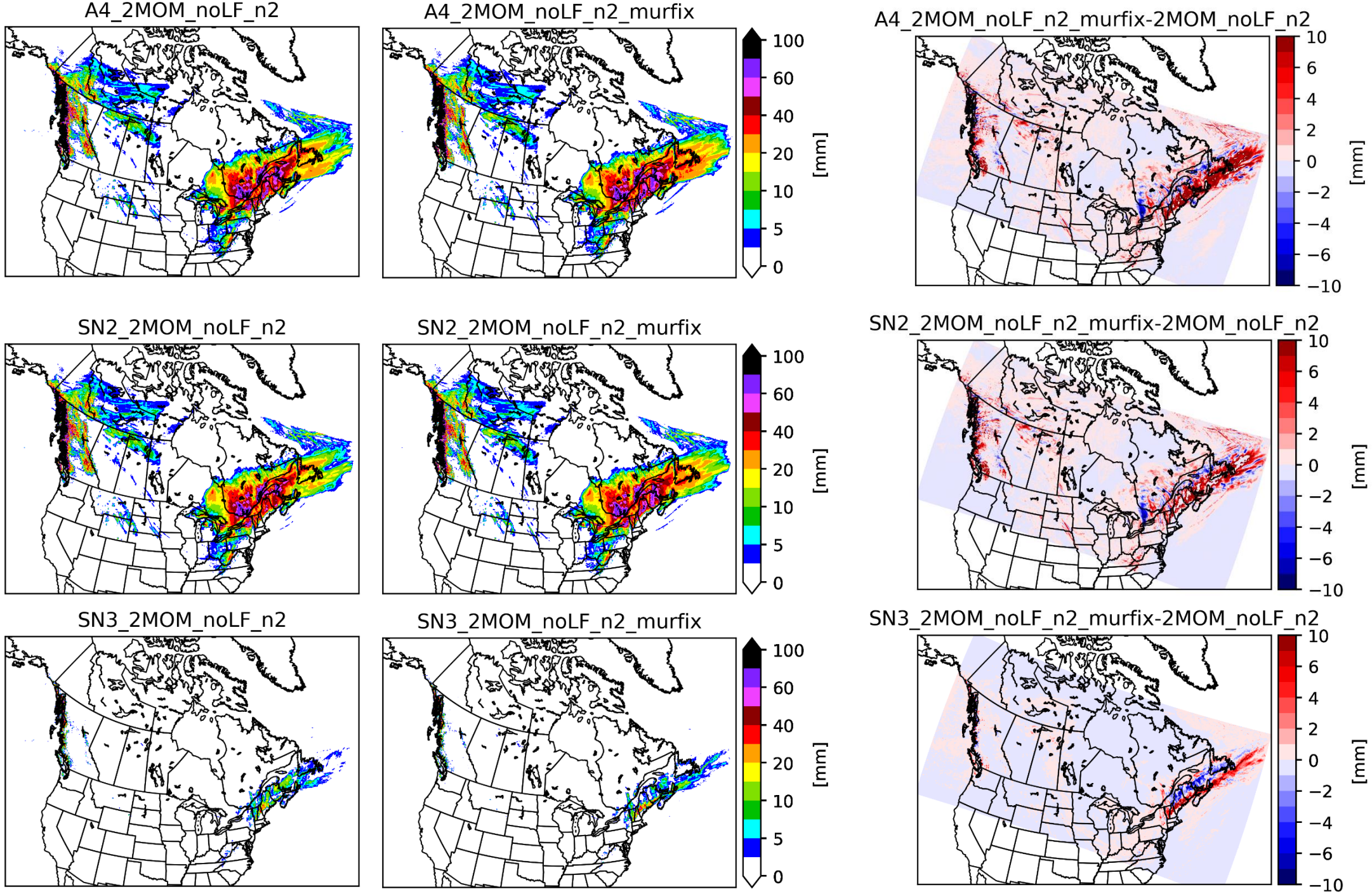
March 2023

2MOM_N2

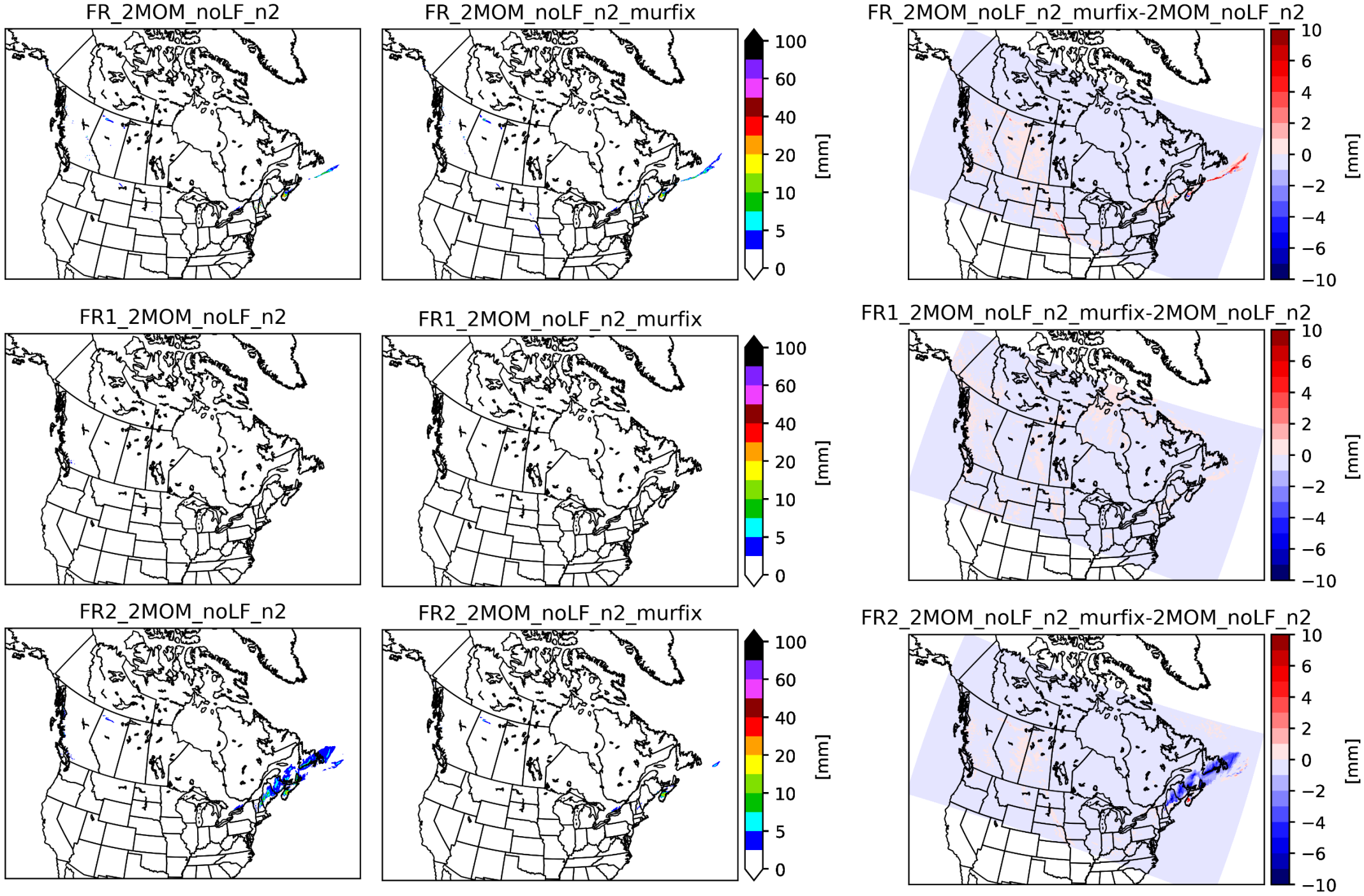
- HRDPS Freezing rain case: 2020022700 (48-h accumulation)



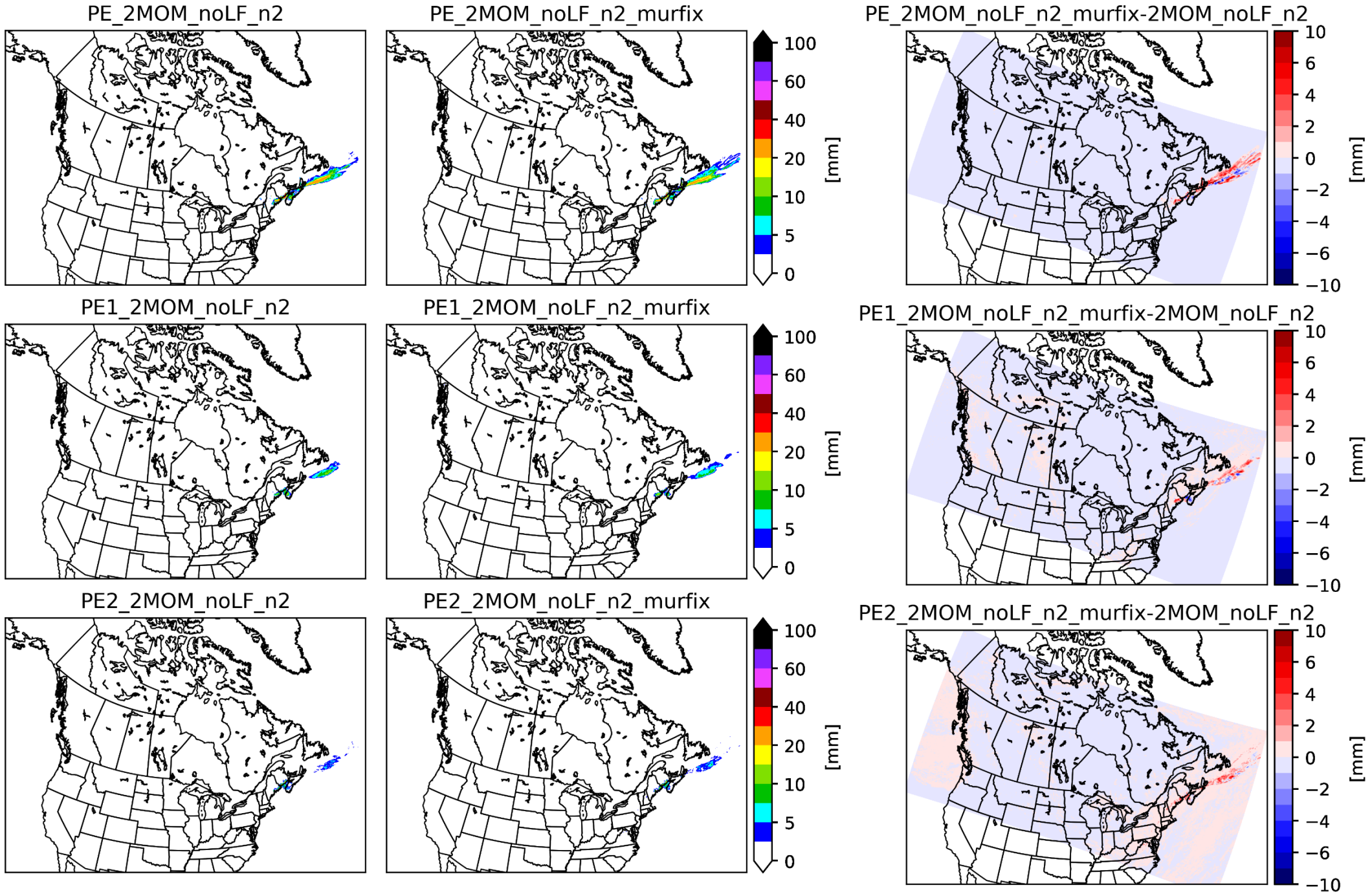
Large increase of PR and SN with nCat=2



Large increase of A4, SN2 and SN3 with nCat=2



Increase in FR, but decrease in FR2



Increase of ice pellets

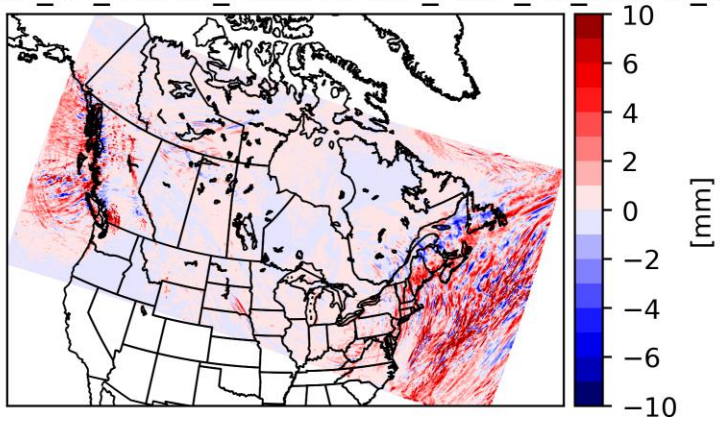
Explications

- Did not find this problem in 1D because we are using prt_liq and prt_sol
- In WRF → should not have this problem
- In GEM → ztss = sn1+sn2+sn3+pe1+pe2 coming from the algo precipitation types.
- When taking ztss = ztss = prt_sol, everything is fine and PR is similar to nCat=1
- The problem is in the diagnostic of ice types:
  - The use of prt_sol(i) in a iice-loop
  - Solution: the use of prt_sol1(i,iice) instead

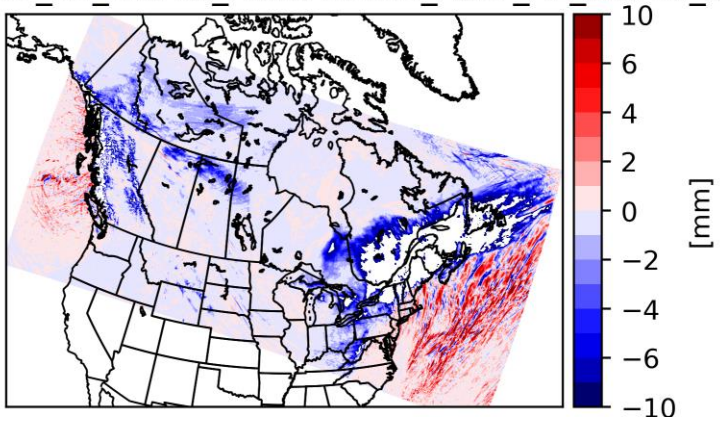
2MOM N2 vs old N2

- Everything looks fine with the accumulation of precipitation types.

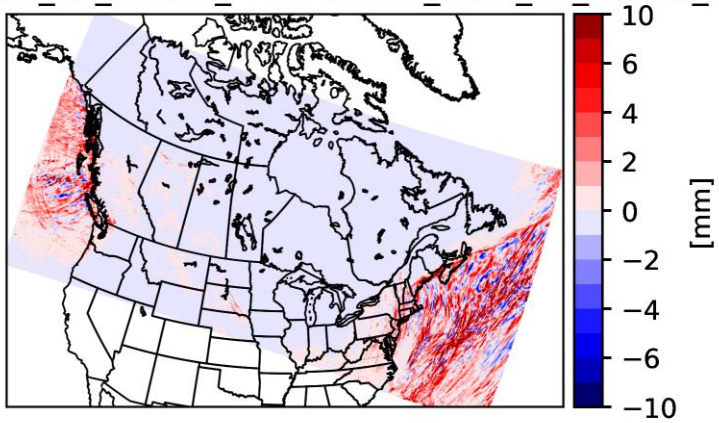
PR_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



PR_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

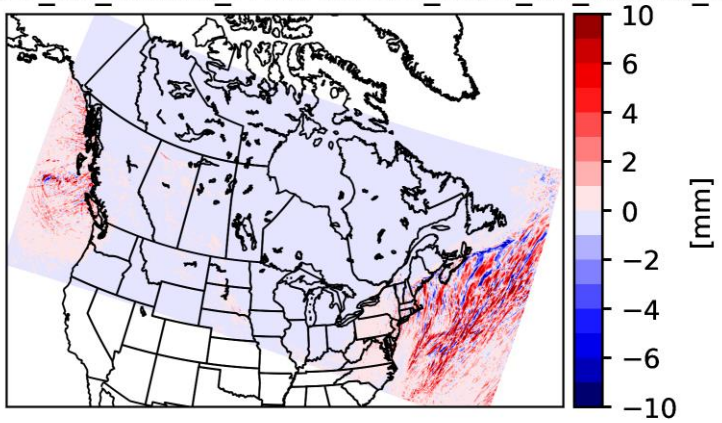


RN_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



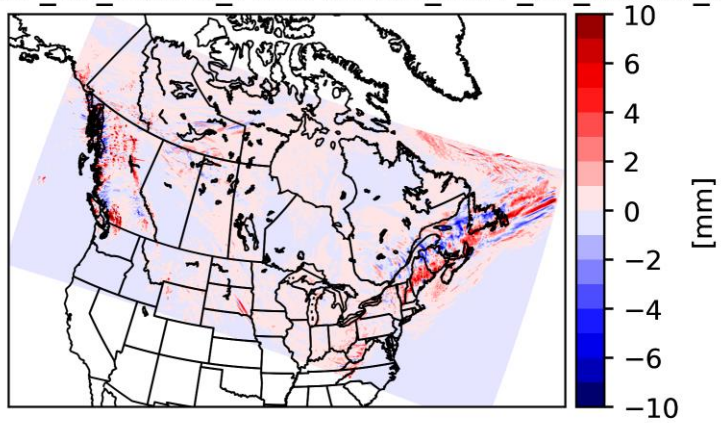
Small increase in PR and RN and SN compared nCat=1

RN_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

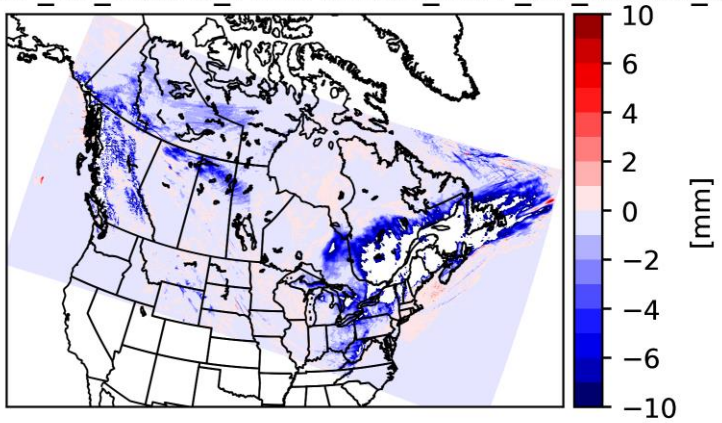


Large decrease of SN and PR as expected compared to old nCat=

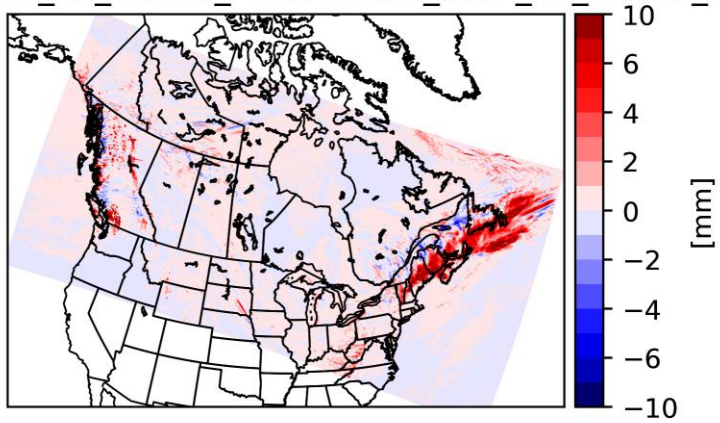
SN_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



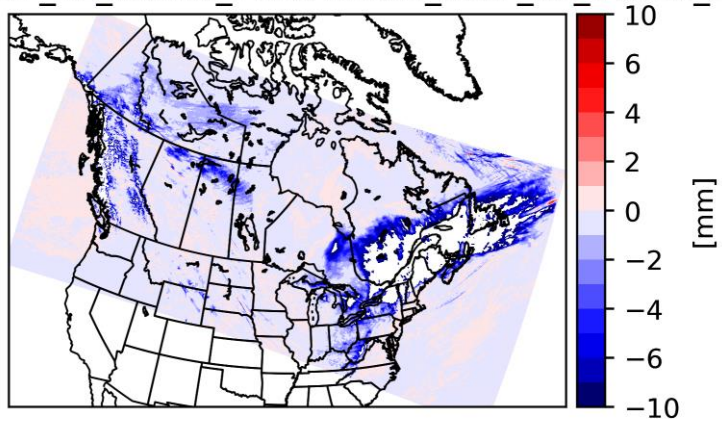
SN_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD



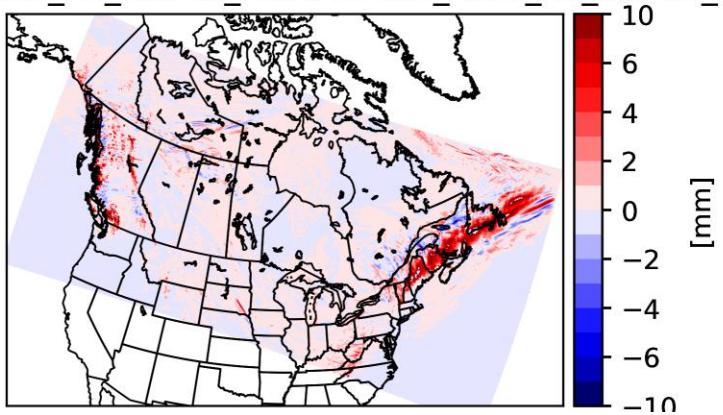
A4_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



A4_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

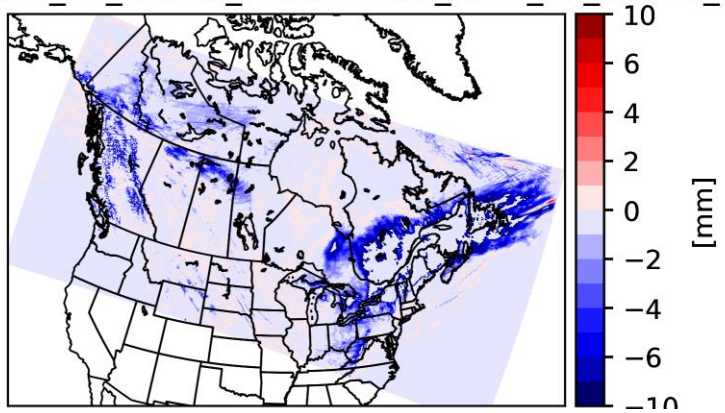


SN2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



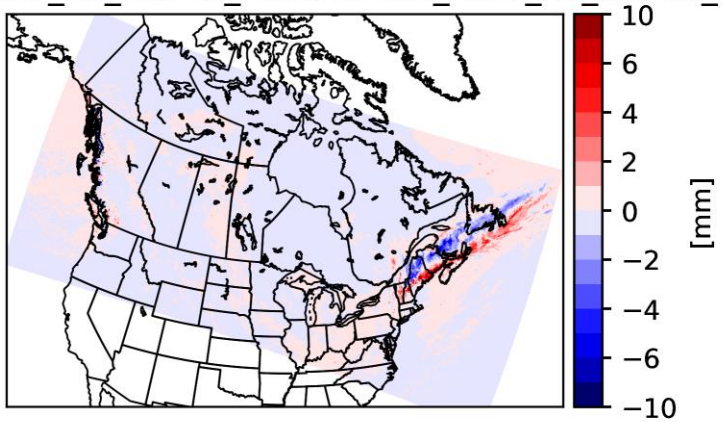
Increase in A4 SN2 and mostly SN3 compared nCat=1

SN2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

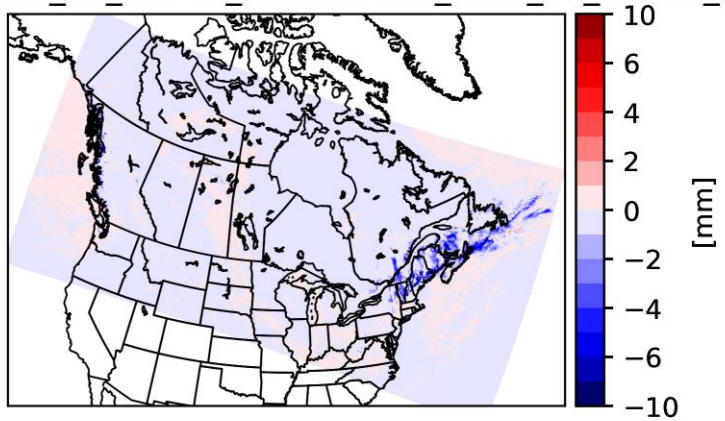


Large decrease of A4, SN2 and SN3 as expected compared to old nCat=2

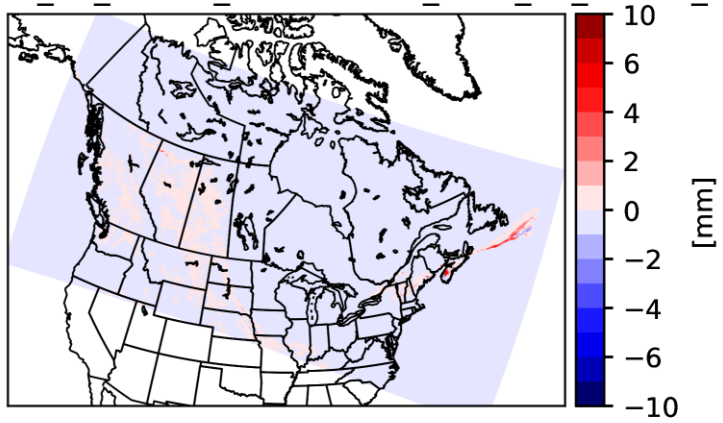
SN3_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



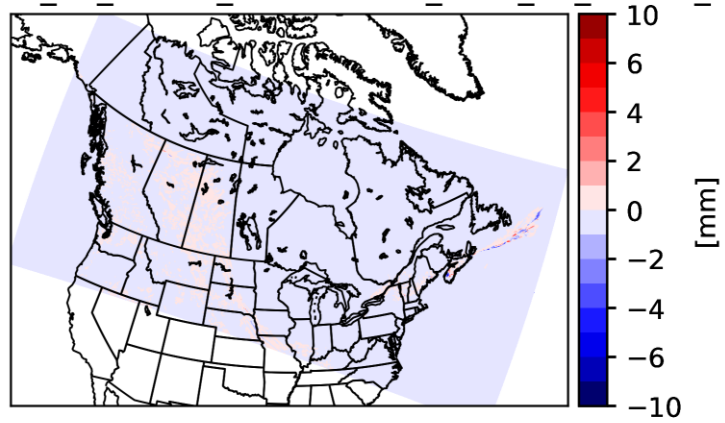
SN3_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD



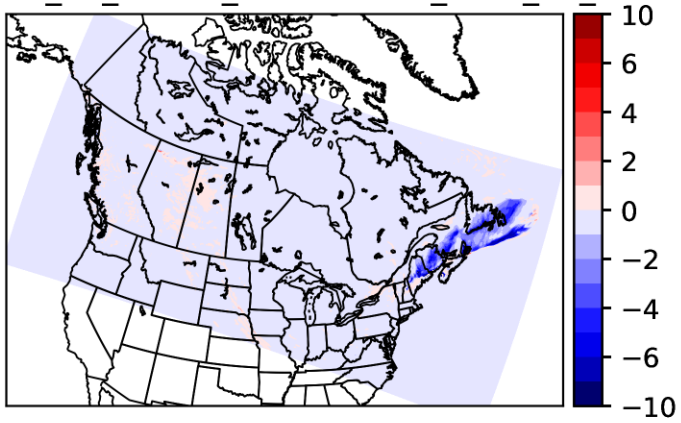
FR_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



FR_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

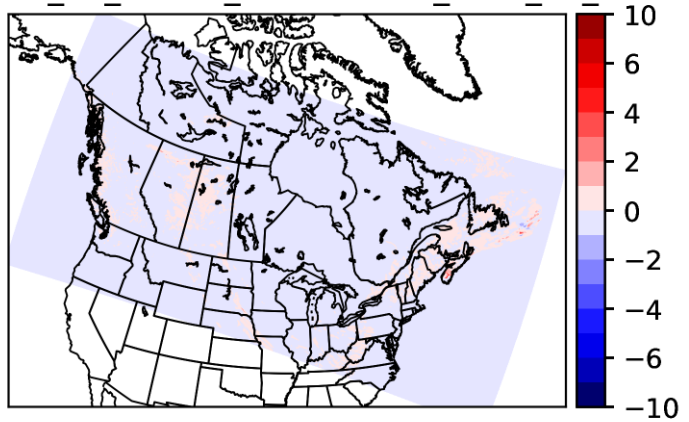


FR2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



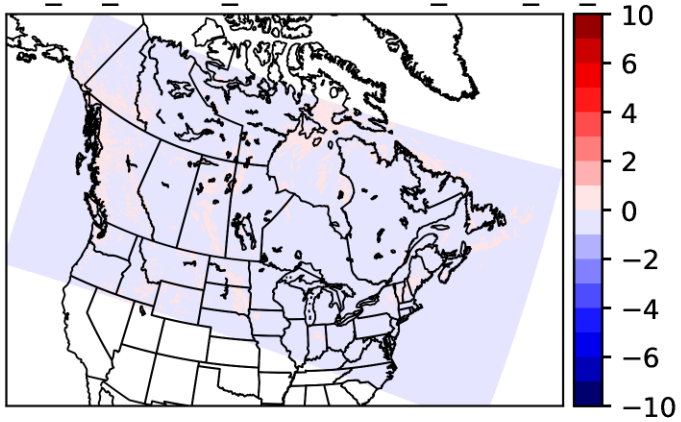
Increase in A4 SN2 and mostly SN3 compared nCat=1

FR2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

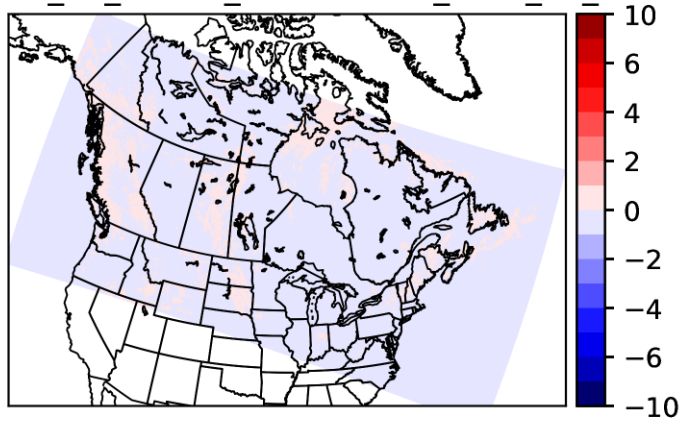


Large decrease of A4, SN2 and SN3 as expected compared to old nCat=2

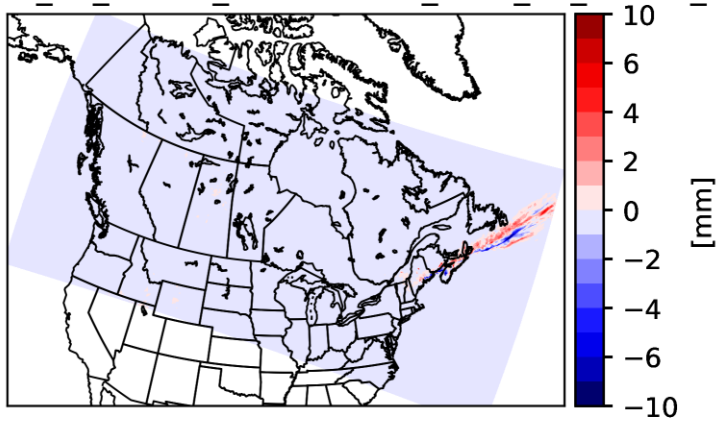
FR1_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



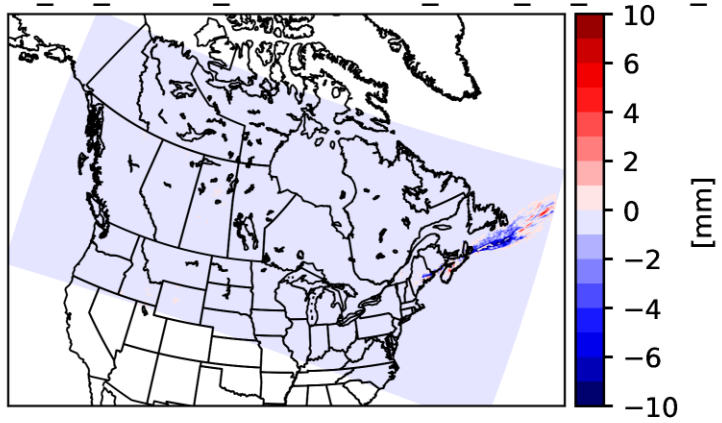
FR1_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD



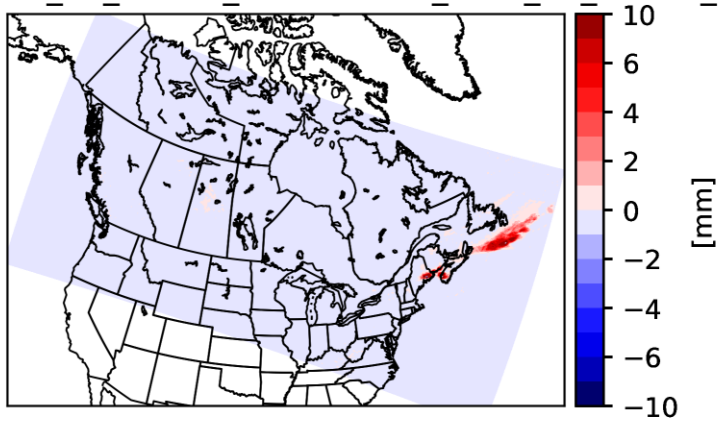
PE_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



PE_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

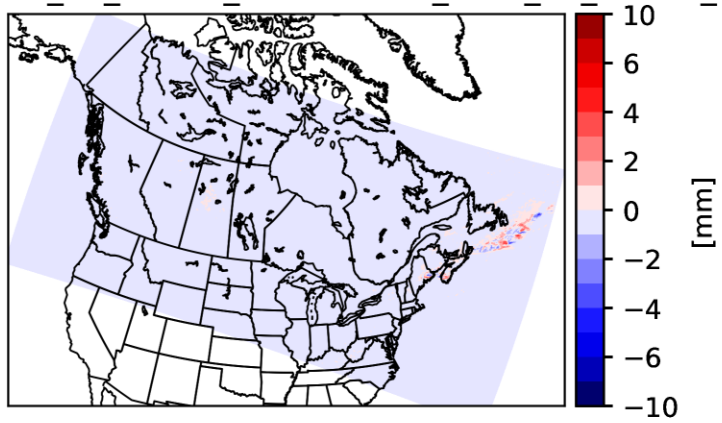


PE1_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



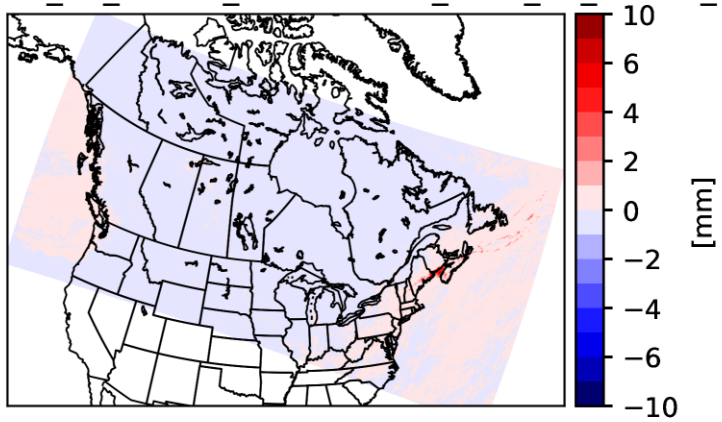
Increase PE, PE1 and PE2 compared to nCat=1 (as expected)

PE1_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD



Decrease of PE and PE2 as expected with the bugfix for nCat=2

PE2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n1_murfix_outD



PE2_2MOM_noLF_n2_murfix_PRfix-2MOM_noLF_n2_murfix_outD

